Wireless Sensor Networks – Lab 1: Cooja simulator

In this first practical lab of the subject we are going to focus on the use of the *Cooja* WSN simulation tool, which is the main Wireless Sensor Network simulator defined within the Contiki OS toolset for resource-constrained devices in Internet of Things. Cooja provides functionalities for simulating and testing the behavior of sensor networks based on modes that run Contiki based applications, starting from simple "hello-word" and "blinking" codes up to advanced multi-hop 6LoWPAN networks with many interconnected wireless nodes.

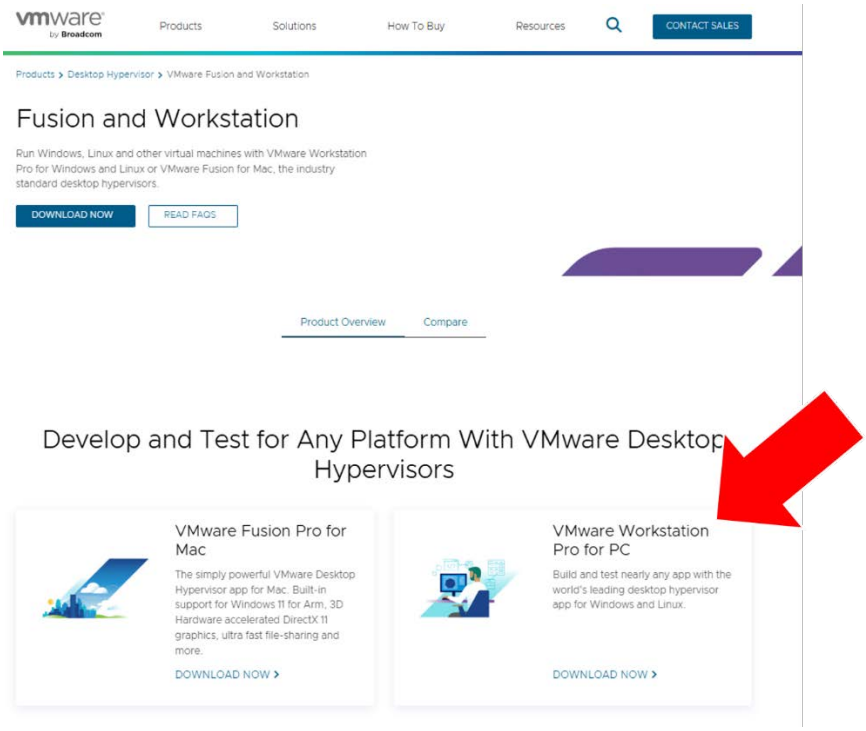
In order to have the Cooja simulation tool properly installed and functional on your computer, we are going to setup in this starting guide a Virtual-Machine-based development environment which will integrate all the files, associated tools and capabilities to have the simulator up and running. To do so, you have to follow step-by-step the next instructions:

1. Installation of the Virtualization Tool

The development environment will run in a Linux operating system. Thus, a virtualization tool will be installed so as to set up the virtual machine that will run the provided development environment. There are several virtualization tools available, but we are going to use the **VMWare Workstation pro** for this purpose.

- Download and install VMWare from the following link, which is available for Windows 64-bit Operating Systems:

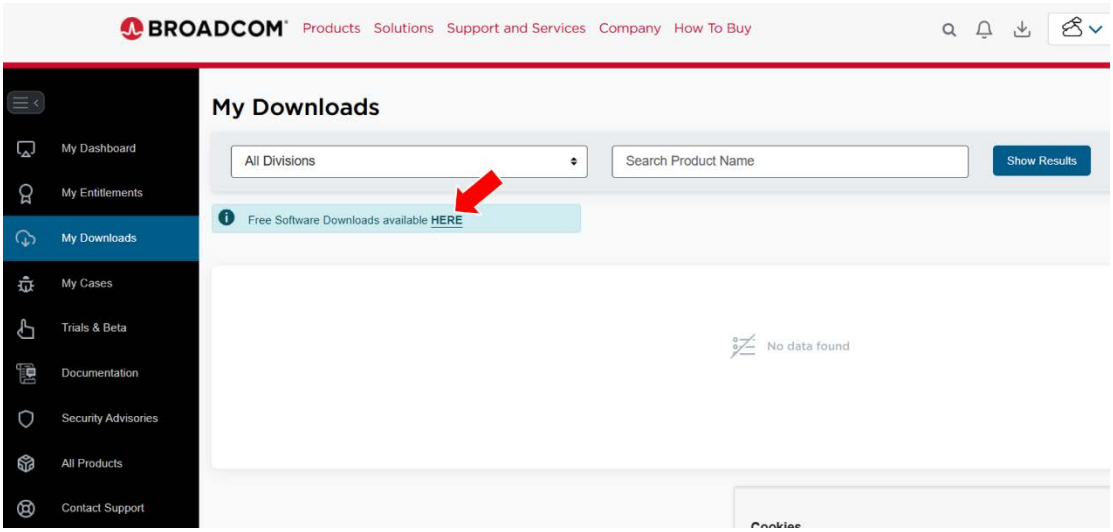
  https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion



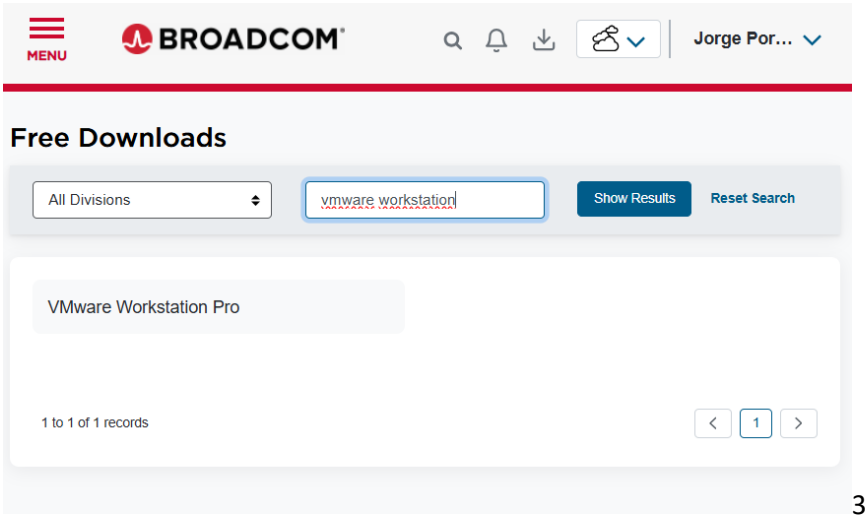
- Login (you need to create an account)



- Go to Free SW

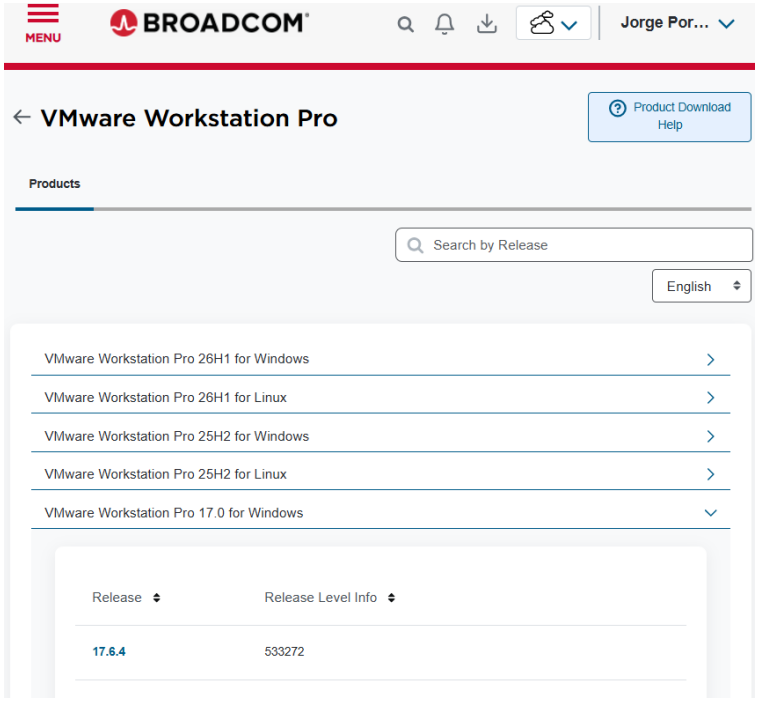


- Look for VMWare Workstation



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- Choose version 17 (and after 17.6.4)
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- Install it.
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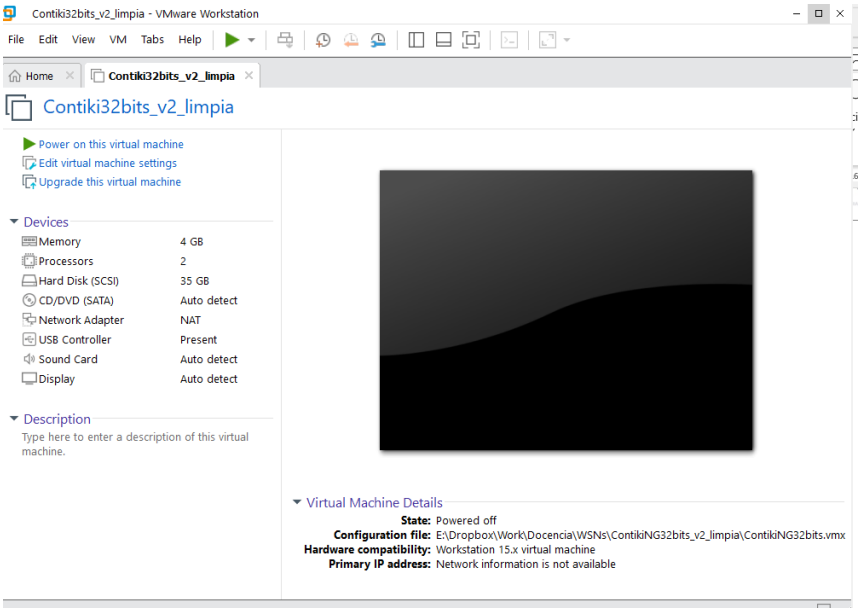
2. Download the Instant Contiki Virtual Machine

The development environment for Cooja is packed into an Ubuntu-based Linux virtual machine that can be downloaded from the following link.

- Download the file from: link
- Unzip the downloaded .zip file in a convenient location, and you will have a folder called *ContikiNG32bits_v2_limpia.*

3. Open the Virtual Machine from the VMWare

Open the VMWare tool, and at the top left-hand side menu called *Player* select *File > Open*



- Go to the folder created in the previous step and select the file: *ContikiNG32bits_v2_limpia.*

- You can edit the configuration of the Virtual Machine by using the option “Edit virtual machine settings” that appears at the bottom right-hand side of the VMWare tool (or by right-clicking on the newly created virtual machine and selecting the option settings…), and modify parameters such as the allocated RAM memory size for the VM, number of processors, hard-disk space, and so on. In case you do not have too much resources on your computer, you can leave it as default.

4. Run the virtual machine

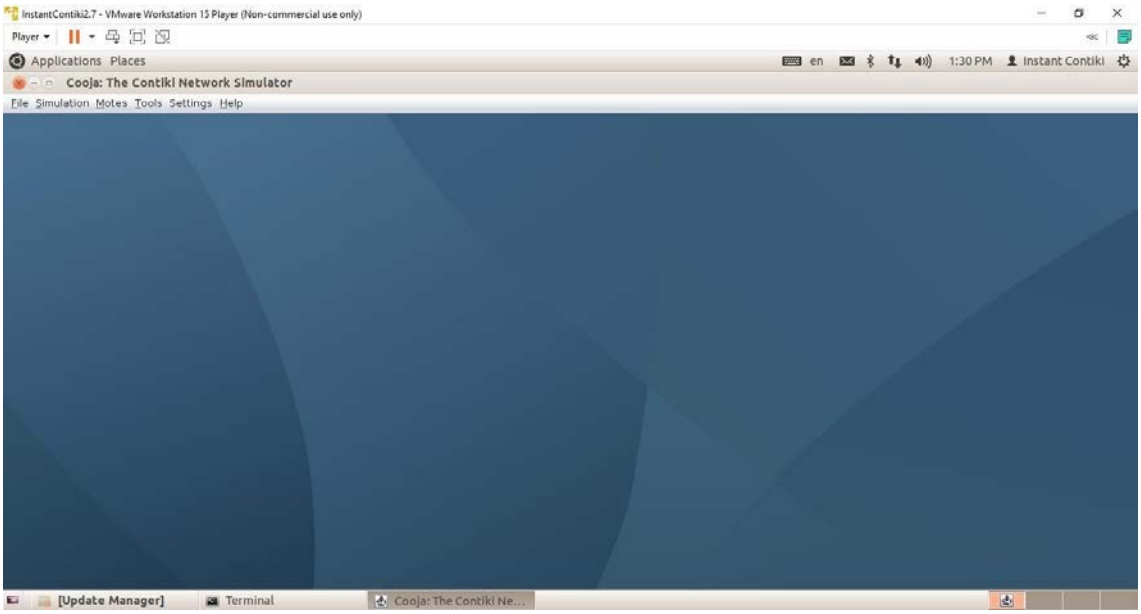
- Click on the green button or double-click on the virtual machine to run it.
- In case you have a message saying that the virtual machine has been moved or copied, select the option “I Copied it”. This message will appear only the first time the VM is run.
- The Ubuntu Operating System will start, and you have to log in. Select the user and tape “user” as password.

NOTE: If you have a message from the VMWare tool at the bottom of the window saying that the VMWare Tools are not up to date, you can follow the steps to update them, so that it can solve minor issues such as not being able to see the Ubuntu OS in full screen.

5. Run Cooja

Once the Ubuntu OS is up and running, you should see the Cooja icon on the Ubuntu OS desktop:

Double click on the Cooja Icon to start the simulator. Wait few seconds until the tool is automatically configured and launched, then you should see the following screen:



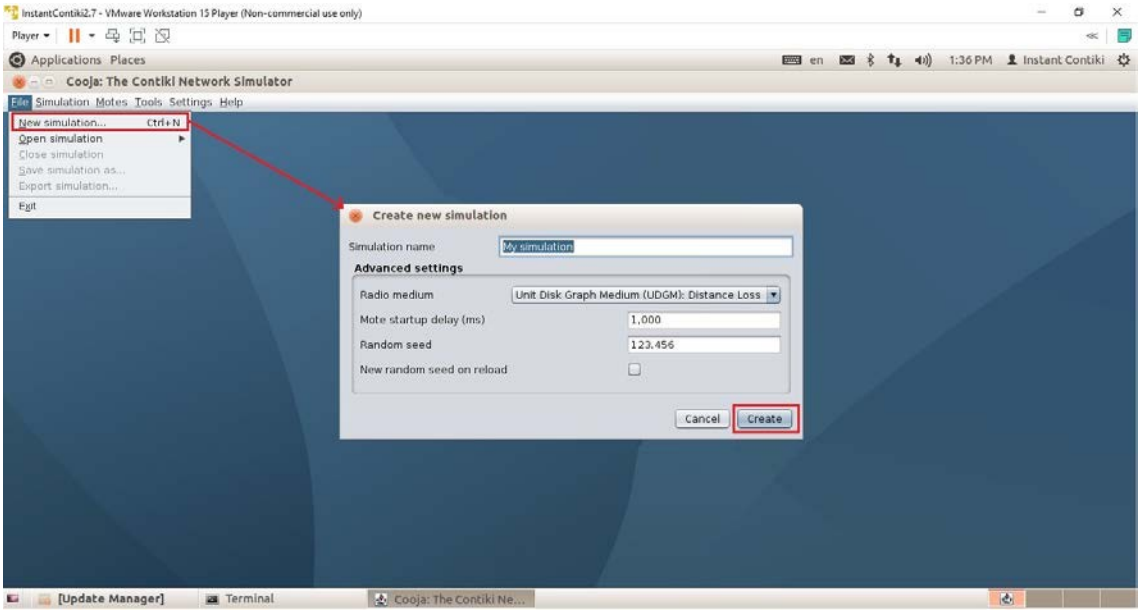
- Now the Cooja network simulator is ready to be used.

6. Verify the simulation environment

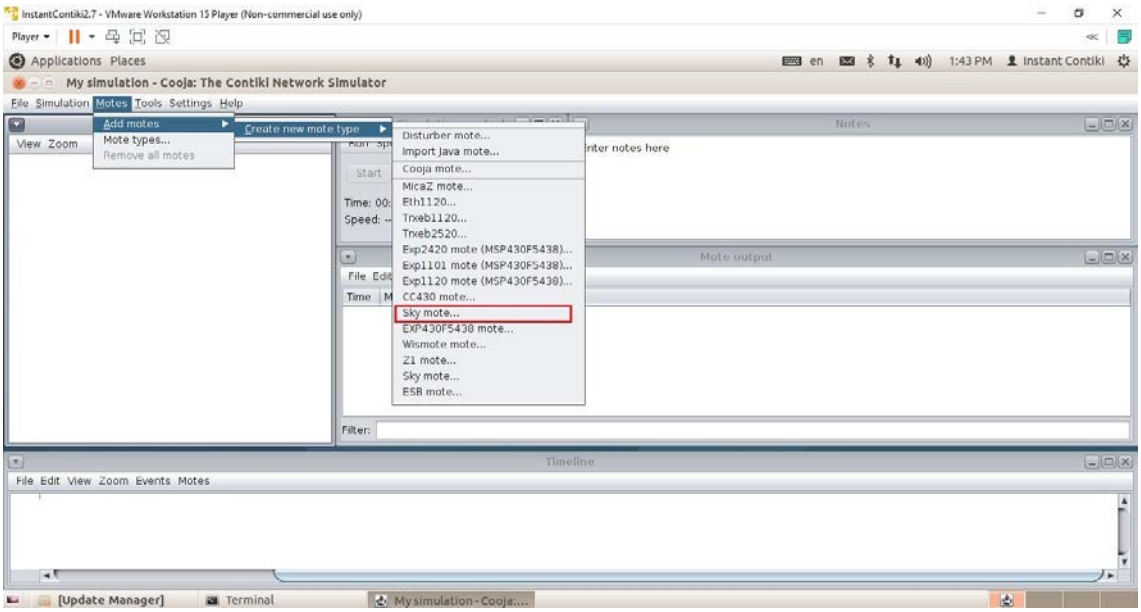
In order to verify and confirm that Cooja is working properly, let’s create and run a first simulation with the tool.

- Click on File > New Simulation.

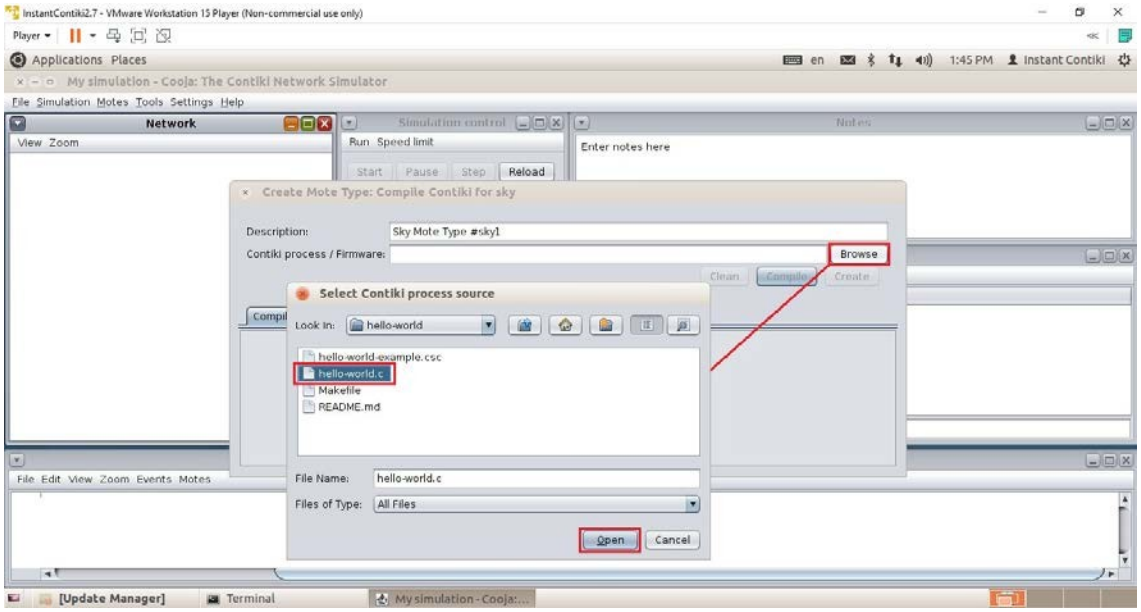
In the new simulation menu, leave the initial settings as default and click on *Create*.



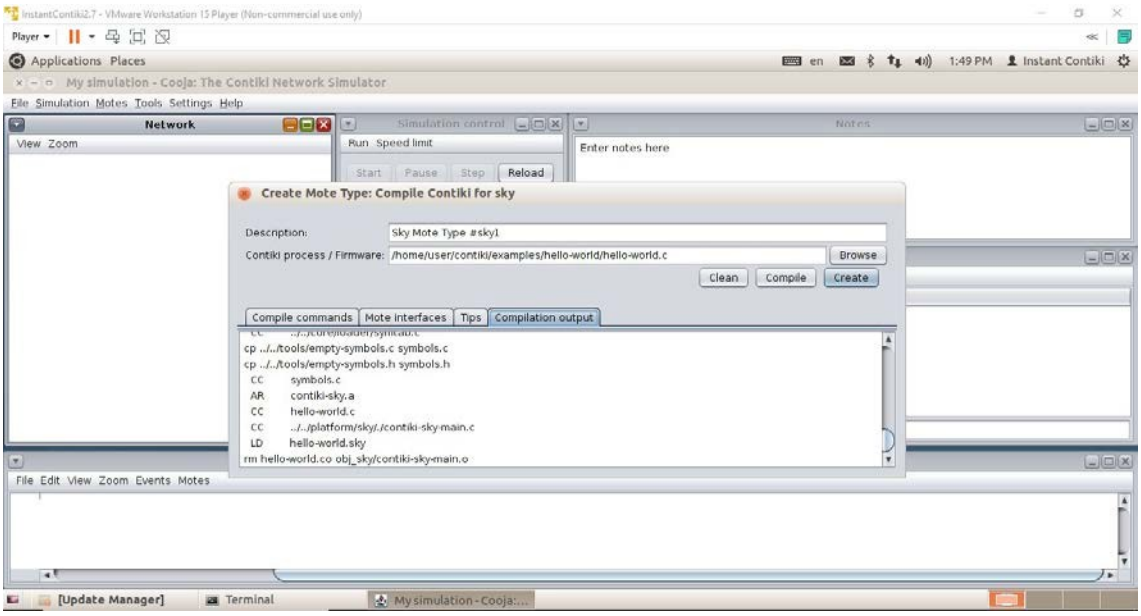
- Go to “Motes > Create New Mote Type” and select “Sky Mote…”



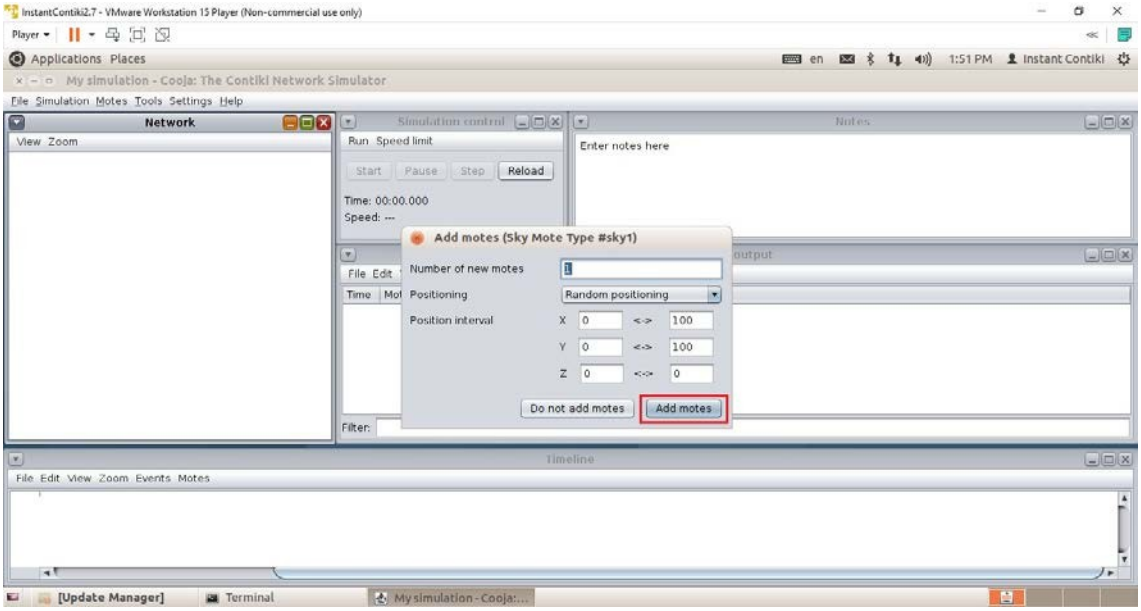
- Then a compilation options menu appears. Click on Browse, and select the “*helloworld.c*” source file, and click on open.



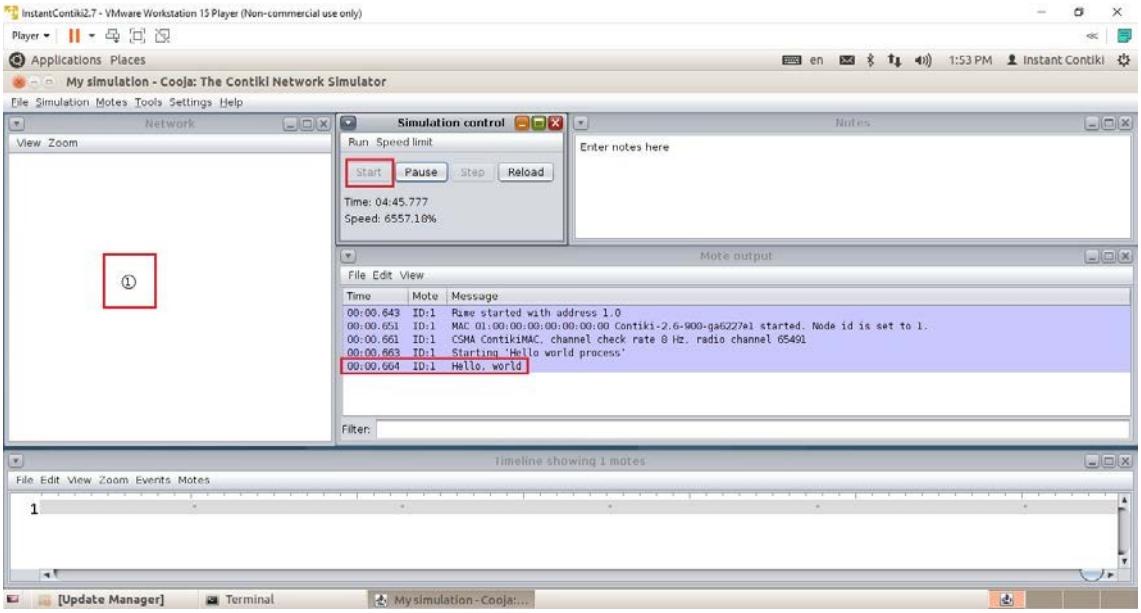
- Click on *Compile*, and once the compilation process is completed, then click on *Create*.



- In the next menu, add only one node and leave the rest as default.



- A node should then appear in the **network view** window. Click on **Start** in the **Simulation control** window and then in the **Mote output** window you should see the “Hello-Word” Message after initial configuration messages.



Now your Cooja Network Simulator environment is ready to be use during the Wireless Sensor Network lab.

- Enjoy the Lab!